Requirements Specification

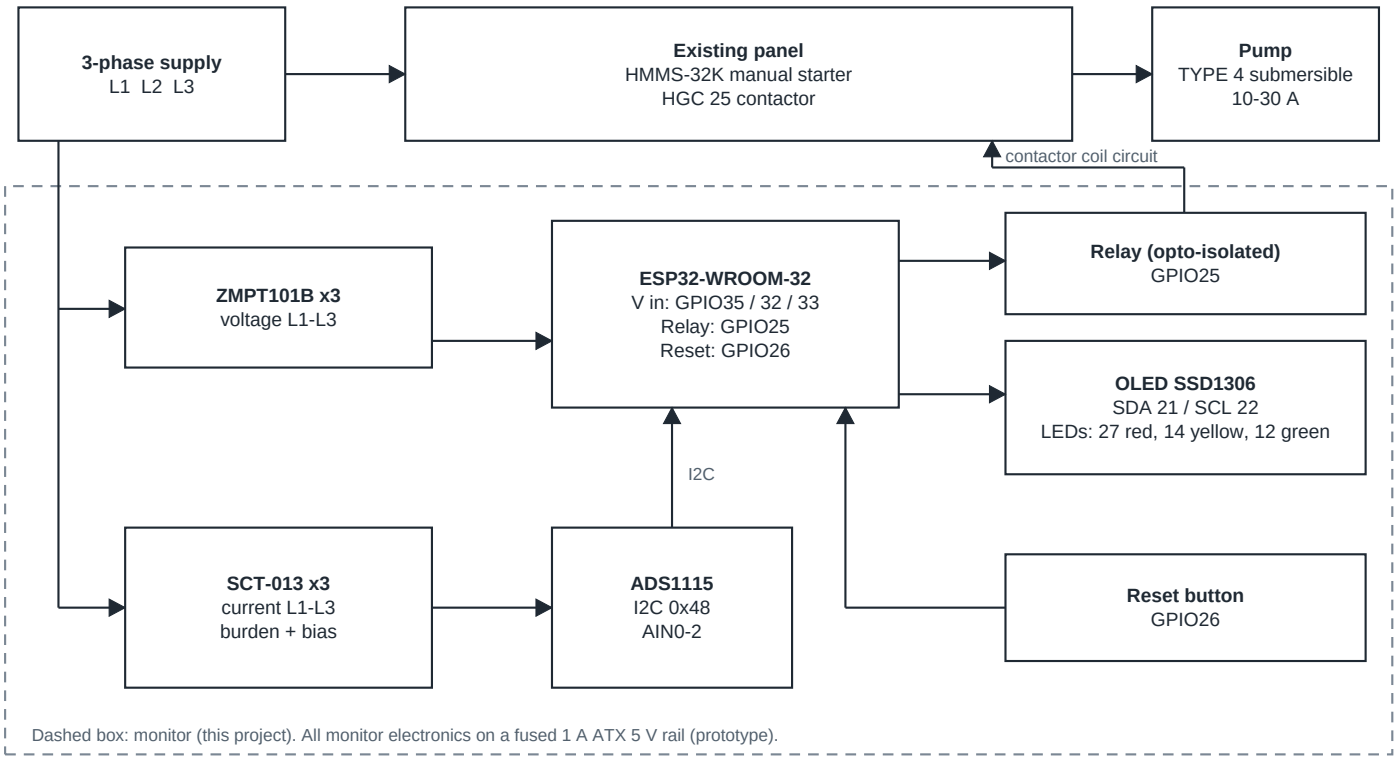
3-phase pump protection monitor (industrial-motor-monitor)

Item	Detail
Document	Requirements Specification, Rev A (draft)
Revision	A, 7 October 2026: initial draft for the industrial-motor-monitor repository.
Date	7 October 2026
Author	Alireza Gholizadeh Vazvani
Basis	Bench prototype status as of the 28 Sep and 3 Oct 2026 progress reports. Firmware source is not yet in this repository.
Validation status	Bench only. Not validated on the real pump panel. Not a certified protection device.

1. Scope

Item	Detail
System	ESP32-based monitor of three phase voltages and three phase currents that opens the contactor coil circuit on a fault.
Target	Jahan Pump Co. TYPE 4 series 3-phase submersible drainage pumps, 10-30 A, on an existing panel with a Hyundai HGC 25 contactor and an HMMS-32K manual motor starter.
Mains frequency	50 Hz assumed in timing analysis.
Out of scope	Replacing the HMMS-32K or HGC 25 protection. Product certification. Remote or network functions.

2. System block diagram



3. Conventions

Term	Meaning
REQ-F / REQ-H / REQ-N	Functional, hardware, and non-functional requirements. IDs are shortened to F-nn, H-nn, N-nn in the tables.
Verif.	T = test, I = inspection, A = analysis, D = demonstration.

Term	Meaning
Bench-checked	Observed working on the breadboard. Not validated on the real panel.
Implemented	Written in firmware per the 3 Oct report. Not yet verified end to end.
Open	Not done, not verified, or not yet defined.

4. Functional requirements

ID	Requirement	Verif.	Status	Notes
REQ-F-01	The system shall measure the RMS voltage of each of the three phases.	T	Implemented	Raw channels steady at idle (28 Sep). Calibration on the real panel pending. See R-02.
REQ-F-02	The system shall measure the RMS current of each of the three phases.	T	Implemented	Calibration on the real panel pending. See R-05.
REQ-F-03	The system shall trip when any phase current exceeds 32 A, outside the startup delay.	T	Implemented	Threshold is adjustable. See R-01.
REQ-F-04	The system shall trip when any phase voltage is outside 180 V to 250 V.	T	Implemented	Measuring reference not yet confirmed. See R-02.
REQ-F-05	For 4 s after the motor starts, the system shall not trip on inrush current.	T	Implemented	Real inrush duration unmeasured. See R-03. Start detection to be documented from the firmware source.
REQ-F-06	The system shall trip on loss of a phase and on phase imbalance above a defined limit.	T	Open	Imbalance limit not yet defined.
REQ-F-07	Protection functions shall be prioritised in this order: over/under-voltage, over-current, phase loss/imbalance.	I	Open	To be confirmed against the firmware source.
REQ-F-08	On any trip the system shall open the contactor coil circuit through the relay on GPIO25.	T	Implemented	Relay not yet wired into the HGC 25 coil circuit.
REQ-F-09	A trip shall latch. Only the manual reset button (GPIO26) shall clear it. The system shall not restart automatically.	T	Implemented	Behaviour after controller power loss: see R-04.
REQ-F-10	The OLED shall show the state OK, STARTING, or TRIPPED.	I	Implemented	
REQ-F-11	The red LED shall be on while the system is tripped.	I	Implemented	Yellow and green LED meanings to be documented.
REQ-F-12	Calibration factors shall be defined in one place at the top of the firmware source.	I	Implemented	

5. Hardware requirements

ID	Requirement	Verif.	Status	Notes
REQ-H-01	The controller shall be an ESP32-WROOM-32.	I	Bench-checked	
REQ-H-02	Three ZMPT101B voltage sensors shall connect to ESP32 ADC pins GPIO35, GPIO32, GPIO33.	I	Bench-checked	Noise handled by software filtering and calibration.
REQ-H-03	Three SCT-013 current sensors shall connect through a burden resistor and bias divider to ADS1115 inputs AIN0-2 at I2C address 0x48.	I	Bench-checked	ADS1115 detected at 0x48. Bias about 1.56 V on AIN0-2 (28 Sep). Sensor variant and burden to be confirmed in hardware.md.
REQ-H-04	The current channels shall not clip at or below the 32 A trip threshold.	A	Open	Needs confirmed sensor variant, burden, and ADS1115 gain.
REQ-H-05	The trip output shall be an opto-isolated relay driven from GPIO25.	I	Bench-checked	Relay module: TONGLING JQC-3FF.
REQ-H-06	The monitor shall be powered from a 5 V rail through a 1 A fuse.	I	Bench-checked	ATX rail in the prototype. Supply for panel installation not yet defined.
REQ-H-07	A 0.96 in SSD1306 OLED shall connect on I2C (SDA 21, SCL 22). LEDs shall connect to GPIO27 (red), GPIO14 (yellow), GPIO12 (green).	I	Bench-checked	

ID	Requirement	Verif.	Status	Notes
REQ-H-08	The monitor shall act only on the contactor coil circuit. The existing HMMS-32K and HGC 25 protection shall remain connected.	I	Open	Pending panel integration.
REQ-H-09	Voltage and current sensor inputs shall be protected against overvoltage and transients.	A	Open	Not present in the prototype. See R-06.
REQ-H-10	The relay state on controller reset, hang, or loss of power shall be defined, documented, and verified. A watchdog shall be enabled.	T	Open	See R-04.
REQ-H-11	For installation on the panel the build shall be insulated and enclosed. The breadboard shall not be connected to the live panel.	I	Open	See safety.md.

6. Non-functional requirements

ID	Requirement	Verif.	Status	Notes
REQ-N-01	Thresholds and calibration constants shall be documented in the repository with their basis.	I	Open	
REQ-N-02	The time from fault to relay open shall be specified and measured.	T	Open	Target value not yet set. Depends on R-05.
REQ-N-03	Each requirement shall trace to a verification method and a recorded result.	I	Open	
REQ-N-04	The firmware shall build from the repository with PlatformIO.	D	Open	Firmware source pending transfer into the repository.
REQ-N-05	The documentation shall state the validation status (bench only) and that the device is not certified.	I	Implemented	docs/safety.md.

7. Risk register

ID	Risk	Effect	Action	Related	Status
R-01	Over-current margin. The 32 A threshold is 2 A (6.7 percent) above the top of the 10-30 A pump range.	A smaller pump can run well above its rating without tripping. A 30 A pump may trip on normal peaks.	Set the threshold per pump from the nameplate current. Measure running current under load. Keep the HMMS-32K overload setting as primary protection.	F-03, H-04	Open
R-02	Voltage window phase reference. 180-250 V matches a phase-to-neutral value near 230 V. Phase-to-phase on this supply reads near 400 V (230 x 1.732 = 398 V).	A wrong reference trips at once or never trips. The panel may have no neutral.	Confirm panel wiring and what each ZMPT101B measures. Define the window per reference. Calibrate against a multimeter.	F-01, F-04	Open
R-03	Startup delay versus real inrush. The 4 s delay is an assumption.	Too short: nuisance trip at start. Too long: no over-current protection during a start.	Record the starting current of the real pump. Set the delay from the measured inrush plus margin.	F-03, F-05	Open
R-04	Fail-safe and watchdog. Relay state on ESP32 hang, reset, or power loss is not verified.	A hung controller may leave the contactor in an unknown state and fail to trip.	Define the relay default state. Test reset and power loss. Enable and test a watchdog.	F-08, F-09, H-10	Open
R-05	ADS1115 sample rate not measured. Datasheet maximum is 860 SPS. Shared by three multiplexed channels: at most 287 SPS each, about 5.7 samples per 50 Hz cycle.	Poor RMS accuracy and slow trip response.	Measure the effective rate in firmware. Compare RMS against a clamp meter. Adjust sampling if needed.	F-02, N-02	Open
R-06	Missing input protection on the sensor inputs.	A transient or wiring fault can damage the ESP32 ADC or ADS1115 inputs.	Add series resistance and clamping. Define and test the protection.	H-09	Open